

# LINEAR REGRESSION

Revision Cheat Sheet

## 1 · WHAT IS LINEAR REGRESSION?

- **Supervised** learning algorithm used to predict a **continuous numerical output**.
- Supervised because during training the model sees **input x** → **known output y** and learns the relationship between them.
- **Simple LR** = one input feature; **Multiple LR** = many input features; **Polynomial LR** = curved relationship.
- **Best Fit Line** = the line that represents the data as accurately as possible — errors are as small as possible.

## 2 · EQUATION OF LINE / PLANE / HYPERPLANE

| #Features | Geometry        | Equation                                 |
|-----------|-----------------|------------------------------------------|
| 1         | Line            | $\hat{y} = w_1x_1 + b$                   |
| 2         | Plane           | $\hat{y} = w_1x_1 + w_2x_2 + b$          |
| 3         | Hyperplane (4D) | $\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + b$ |
| n         | Hyperplane      | $\hat{y} = w_1x_1 + \dots + w_nx_n + b$  |

- In ML we write  $\hat{y} = mx + b$  — **m (weight)**: change in output per 1-unit change in input; **b (bias)**: offset.

Prediction = **Effect of Input** + **Offset**

## 3 · DISTANCE OF A POINT FROM A PLANE

- Plane:  $Ax + By + Cz + D = 0$  (A,B,C orient the plane; D fixes its position).

$$d = |Ax_0 + By_0 + Cz_0 + D| / \sqrt{A^2 + B^2 + C^2}$$

## 3B · SIMPLE VS MULTIPLE VS POLYNOMIAL

| Type          | Inputs           | Shape learned      |
|---------------|------------------|--------------------|
| Simple LR     | 1 feature        | Straight line      |
| Multiple LR   | 2+ features      | Plane / hyperplane |
| Polynomial LR | 1+ (with powers) | Curve              |

## 7 · REGRESSION METRICS

| Metric         | Formula                          | Note                          |
|----------------|----------------------------------|-------------------------------|
| MAE            | $(1/n) \sum  y_i - \hat{y}_i $   | Same unit; robust to outliers |
| MSE            | $(1/n) \sum (y_i - \hat{y}_i)^2$ | Penalizes large errors        |
| RMSE           | $\sqrt{\text{MSE}}$              | Same unit; penalizes outliers |
| R <sup>2</sup> | $1 - \text{SSE}/\text{SST}$      | Variance explained            |

- Smaller MAE/MSE/RMSE → **better**; R<sup>2</sup> closer to 1 → **better**.

Example — errors [2,-2,3]: MAE=2.33, MSE=5.67, RMSE=2.38.

## 8 · R<sup>2</sup> SCORE (COEFFICIENT OF DETERMINATION)

- Asks: how much variation in the target does my model explain, vs. just predicting the mean?

$$\text{SST} = \sum (y_i - \bar{y})^2 \quad \text{SSE} = \sum (y_i - \hat{y}_i)^2$$

$$R^2 = 1 - \text{SSE}/\text{SST}$$

| R <sup>2</sup> value | Meaning                      |
|----------------------|------------------------------|
| = 1                  | Perfect fit (SSE = 0)        |
| = 0                  | No better than mean baseline |
| < 0                  | Worse than mean baseline     |

## 9 · ADJUSTED R<sup>2</sup>

- Plain R<sup>2</sup> **never decreases** when adding features — even useless ones. Adjusted R<sup>2</sup> penalizes irrelevant predictors.

$$\text{Adj. } R^2 = 1 - (1 - R^2)(n - 1) / (n - p - 1)$$

- n = #observations, p = #predictors. Feature helps ⇒ Adj. R<sup>2</sup> ↑; feature is noise ⇒ Adj. R<sup>2</sup> ↓.

## 9B · MAE VS MSE VS RMSE VS R<sup>2</sup>

## 4 · FINDING M AND B

### Two solution paths

| Closed-Form                  | Non-Closed-Form  |
|------------------------------|------------------|
| Ordinary Least Squares (OLS) | Gradient Descent |

## 5 · DERIVING OLS (SIMPLE LR)

- Error for point i:  $d_i = y_i - \hat{y}_i$ , and  $\hat{y}_i = mx_i + b$ .
- Errors can be negative, so square them:

$$E(m, b) = \sum (y_i - mx_i - b)^2$$

- **Goal**: minimize E over m and b ⇒ set  $\partial E / \partial b = 0$  and  $\partial E / \partial m = 0$ .
- Solving  $\partial E / \partial b = 0$  gives: **b =  $\bar{y} - m\bar{x}$**

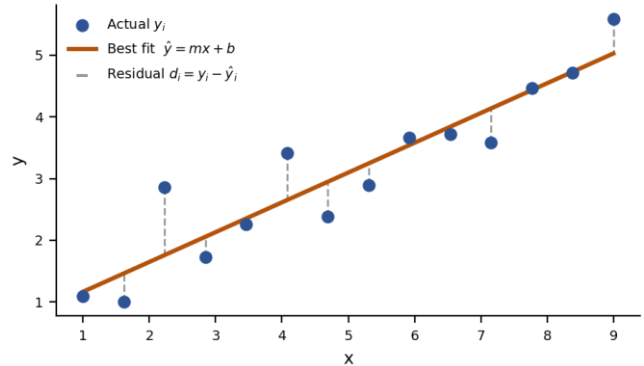
## 6 · FINAL OLS FORMULAS

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b = \bar{y} - m\bar{x}$$

- These are exactly what `MeraLR.fit()` computes from scratch, matching `sklearn.LinearRegression`.

Least Squares: minimize  $E = \sum d_i^2$



## 6B · WORKED NUMERIC EXAMPLE (TITANIC-STYLE)

- Placement data: `cgpa` → `package`. From scratch fit: **m ≈ 0.558**, **b ≈ -0.896**. Predict for `cgpa` = 8.58:  $\hat{y} = 0.558(8.58) - 0.896 \approx 3.89$
- Matches `sklearn's LinearRegression()` exactly — OLS has one unique closed-form solution.

## 10 · MULTIPLE LINEAR REGRESSION (MLR)

- LR with **more than one** input feature predicting one continuous target.

$$\hat{y} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_nx_n$$

- $\beta_0$  = intercept/bias;  $\beta_j$  = weight of feature  $x_j$ , learned from training data.
- Comparing  $\beta_1$  vs  $\beta_2$  directly is **not valid** — magnitude depends on each feature's scale/units.

## 11 · MATRIX (NORMAL EQUATION) FORM

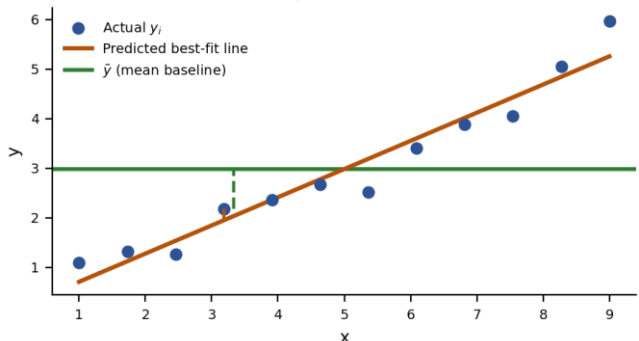
$$\hat{y} = X\beta, \quad X \text{ has a leading column of } 1\text{'s (for } \beta_0\text{)}$$

- Minimizing  $E = (y - X\beta)^T (y - X\beta)$  and setting  $\partial E / \partial \beta = 0$  gives the closed-form solution:

$$\beta = (X^T X)^{-1} X^T y$$

- This is exactly what `MeraLR` (multiple) computes via `np.linalg.inv` — matches `sklearn's coef_/intercept_` exactly.

R<sup>2</sup> : variance explained vs mean baseline



- **MAE, MSE, RMSE** → "How wrong are my predictions?" (error magnitude).
- **R<sup>2</sup>, Adjusted R<sup>2</sup>** → "How well does the model explain the target?"
- Rule of thumb: report **RMSE** for interpretability + **R<sup>2</sup>/Adj. R<sup>2</sup>** for explanatory power.

12 · MAIN ASSUMPTIONS OF LR

| Assumption           | Simple meaning                       |
|----------------------|--------------------------------------|
| Linear relationship  | Inputs & target are linearly related |
| No multicollinearity | Predictors aren't highly correlated  |
| Normal residuals     | Errors ≈ normal distribution         |
| Homoscedasticity     | Residual variance stays constant     |
| No autocorrelation   | Errors aren't related to each other  |

13 · QUICK FORMULA RECAP

| Concept                 | Formula                                                                  |
|-------------------------|--------------------------------------------------------------------------|
| Simple LR               | $\hat{y} = mx + b$                                                       |
| OLS slope               | $m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$ |
| OLS intercept           | $b = \bar{y} - m\bar{x}$                                                 |
| MLR (matrix)            | $\hat{y} = X\beta$                                                       |
| Normal equation         | $\beta = (X^T X)^{-1} X^T y$                                             |
| R <sup>2</sup>          | $1 - \text{SSE} / \text{SST}$                                            |
| Adjusted R <sup>2</sup> | $1 - (1 - R^2)(n - 1) / (n - p - 1)$                                     |